

Demo to production, without rewriting from scratch.

Ten probes of empirical evidence that **extracting the spec** from a working demo is the hard part — and it's tractable.

A great demo is the worst brief.

- The prototype **works** — and customers want it.
- Every quirk, layout choice, and business rule is already decided in code.
- Nobody wrote the spec. The code **is** the spec.
- Hand-rewrites lose intent and cost months.



MONTHS OF ENG TIME, LOST DESIGN INTENT, AND A
REWRITE THAT DOESN'T QUITE MATCH THE DEMO THAT WON
THE DEAL.

You can't write the spec by hand — the demo already **is** one.

The prototype encodes hundreds of decisions: status-pill colors, filter semantics, empty-state copy, the exact 8-column table schema, the waterfall math on the balance cards.

A human spec captures maybe 10%. The rest shows up as "bug reports" months later — when it's the **rewrite** that's wrong, not the demo.

Extracting the spec is hard. Rebuilding from a good spec is **tractable**.

Conventional wisdom: "code generation is the hard part."

We invert this. Modern LLMs can rebuild competently from a clean spec — if you give them one. The work is turning the messy demo **into** that spec.

Two stages.

1 · Reverse-spec-extraction

Four probe agents walk the demo in parallel:

- `ui-walker` — clicks every screen
- `static` — reads the code
- `data` — pulls sample rows & PDFs
- `external` — maps API calls

A reconciler merges them into an **artifact pack** — 6 structured files + screenshots.

2 · Ralph-loop build

A build-controller slices the work by **user journey** and dispatches per-slice executors.

Each slice is verified by:

- oracle test suite
- Playwright journey walks
- **side-by-side screenshot check** against the prototype

Ten probes. Convergence.

PROBE	WALL-CLOCK	WHAT WAS TESTED	KEY OUTCOME
0	—	Manual end-to-end ingest	Pipeline shape validated
1	1h 24m	Automated ingest	Fan-out dispatch + schemas prevent thin merges
1.5	59m	Controller agent (Track 2)	Autonomous orchestration validated
2	55m	First Ralph-loop build	4.5K lines, 45 endpoints, 11 pages — zero human help
2.1	5h 16m	Build-controller + slices	API 122/122; frontend still a shell
2.2	2h 5m	Playwright journey tests	28/29 fail on selector mismatch; orphan pages
2.3	1h 58m	Journey-based slicing	Orphans fixed; dialog theater exposed
2.4	4h 16m	Sample-data + screenshot contract	Gates absorbed; self-evaluation is the structural cause
2.5	4h 27m	Generator-evaluator (plan-reviewer)	Validated — reviewer caught 4 failures; 14/14 journeys · 145/145 oracle

THE MOMENT OF PROOF

Same structure. Real data.

REFERENCE · prototype AR page on empty tenant (all US\$ 0)

07 • PROOF

We learn the system by breaking it.

Orphan pages (2.2)

Slicing by database table left the cross-cutting pages (Dashboard, Analytics) as scaffolds. **Fix:** slice by user journey instead.

Dialog theater (2.3)

Buttons opened dialogs that closed without doing anything. **Fix:** every task must assert **post-action state**, not just "dialog appears."

Empty-tenant blind spot (2.3)

The build never rendered a populated row. **Fix:** seed the prototype's own data before journey walks.

Self-evaluation (2.4 → 2.5)

When the same agent plans the work **and** judges its own plan, it passes every gate while cutting scope. **Fix:** separate author from judge. **Validated in 2.5 — reviewer caught 4 plan-level failures.**

A fast path from prototype to product.

Keep momentum

Prototype in days, productize in days. Design intent survives — because the prototype **is** the spec.

Repeatable

The method works per-probe, so it works per-prototype. Each new Calibrax demo plugs into the same pipeline.

Disciplined

Every failure becomes a probe, a fix, and a piece of institutional memory — not a regression.

The prototype isn't a deliverable anymore. It's **upstream of the product.**

From rewrite tool to **rewrite-and-harden** tool.

JUST COMPLETED • PROBE 2.5

Generator-evaluator validated.

`plan-reviewer` caught 4 plan-level failures across 2 cycles: S08 bundle, deferred-action Uploads, scope leak, consistency drift.

Build shipped clean: **14/14**

NEAR-TERM • THREE TRACKS

Probe 2.6 • interaction fidelity

Static UI handler extraction + reconciler journey-interactions merge. Next probe, already queued.

App shell scaffold

Pre-built monorepo: shared components, schema

BIGGER SHIFT

Constrained artifact improvement loop

Between extract and build, insert specialized hardeners:

`data`, `api-envelope`, `a11y`,
`missing-index`, `error-boundary`.

v1 — constrained: same behavior as demo, but hardened. v2 — **strictly better than the demo**.

The pattern works. Help us compound it.

- **A second prototype to probe against.** Probe-2.5 proved the loop converges on one demo. Generalization is the next thesis — pick a Calibrax prototype with a real customer story.
- **30 minutes with a PM** who's lived a demo → production rewrite. Where did **your** spec leak?

This is no longer a hypothesis under test. It's the first validated architectural change of the probe series.

— PROBE 2.5 • 2026-04-20

- **Resources for the roadmap.** Shell + multi-

THANK YOU

Questions, objections, prototypes.

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